

CODEALPHA
PYTHON PROGRAMMING INTERNSHIP
PROJECT REPORT

TASK 4: BASIC CHATBOT

A Simple Rule-Based Chatbot Using Python

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Python Programming

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BASIC SMART CHATBOT

DECLARATION

I hereby declare that the project entitled “**Basic Chatbot**” has been developed by me as part of the **CodeAlpha Python Programming Internship – Task 3**.

This project was developed using Python programming concepts and follows the requirements provided for the internship task. The application is a simple rule-based chatbot that accepts user input and provides predefined responses.

I declare that the work presented in this report is my own work carried out during the internship period for educational and learning purposes.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **CodeAlpha** for providing me with the opportunity to participate in the Python Programming Internship.

I am thankful for the practical tasks provided during the internship, which helped me apply Python programming concepts to develop useful applications.

I would also like to thank my teachers, mentors, friends, and family for their guidance, encouragement, and support throughout the development of this project.

The Basic Chatbot project helped me improve my understanding of Python functions, conditional statements, loops, input/output operations, and rule-based decision-making.

Finally, I complete this project and I gained new knowledge and experience etc.

ABSTRACT

The **Basic Chatbot** is a simple rule-based conversational program developed using Python as part of the **CodeAlpha Python Programming Internship – Task 3**.

The main objective of this project is to create a chatbot that accepts input from the user and provides predefined responses based on the entered message.

The chatbot recognizes simple inputs such as “**hello**”, “**how are you**”, and “**bye**”. Based on the user's input, the program provides predefined replies such as “**Hi!**”, “**I'm fine, thanks!**”, and “**Goodbye!**”.

The chatbot uses fundamental Python concepts including if-elif conditional statements, functions, loops, variables, and input/output operations.

This project demonstrates how basic programming logic can be used to create a simple interactive conversational application without using artificial intelligence, machine learning, external APIs, or databases.

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1. INTRODUCTION

A chatbot is a computer program designed to interact with users through conversations. Chatbots can receive messages from users and provide responses based on predefined rules or more advanced artificial intelligence techniques.

The Basic Chatbot developed in this project is a simple **rule-based chatbot** created using Python. It accepts text input from the user and compares the input with predefined conditions.

For example, when the user enters **“hello”**, the chatbot responds with **“Hi!”**. When the user enters **“how are you”**, the chatbot responds with **“I'm fine, thanks!”**. When the user enters **“bye”**, the chatbot responds with **“Goodbye!”** and the conversation can be terminated.

The project focuses on fundamental Python programming concepts and provides a simple introduction to conversational program development.

2. PROBLEM STATEMENT

The objective of this project is to develop a simple rule-based chatbot using Python.

The chatbot should accept input from the user and provide predefined responses according to the entered message.

The program should recognize basic inputs such as:

- “hello”
- “how are you”
- “bye”

The chatbot should respond with predefined messages such as:

- “Hi!”
- “I'm fine, thanks!”
- “Goodbye!”

The project should demonstrate the use of if-elif statements, functions, loops, and basic input/output operations.

3. OBJECTIVES

The main objectives of the Basic Chatbot are:

1. To develop a simple rule-based chatbot using Python.
2. To accept text input from the user.
3. To identify predefined user messages.
4. To provide appropriate predefined responses.
5. To use if-elif statements for decision-making.
6. To use functions for organizing chatbot logic.
7. To use loops for continuous conversation.
8. To demonstrate Python input and output operations.
9. To understand the basic logic behind conversational programs.
10. To improve programming and problem-solving skills.

4. SCOPE OF THE PROJECT

The scope of this project is to develop a basic rule-based chatbot that can interact with a user through text input.

The chatbot is designed to recognize a small number of predefined messages and provide corresponding predefined responses.

The current version is intentionally simple and does not use artificial intelligence, machine learning, natural language processing, external APIs, or databases.

The project provides a foundation for understanding how user input can be processed using programming logic and how simple conversational applications can be developed using Python.

5. TECHNOLOGIES AND TOOLS USED

5.1 Python

Python is the main programming language used to develop the Basic Chatbot.

5.2 Conditional Statements

if-elif statements are used to compare the user's input with predefined messages and select an appropriate response.

5.3 Functions

Functions are used to organize the chatbot's response logic and make the program easier to understand and maintain.

5.4 Loops

A loop is used to allow the chatbot to continue interacting with the user until the conversation is ended.

5.5 Visual Studio Code

Visual Studio Code can be used to write, edit, and execute the Python chatbot program.

5.6 Console

The chatbot uses basic console input and output for communication with the user.

6. SYSTEM REQUIREMENTS

6.1 Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM
- Keyboard
- Basic processor
- Sufficient storage space

6.2 Software Requirements

- Windows operating system

- Python 3.x
- Visual Studio Code or another Python IDE
- Python interpreter

6.3 Additional Requirements

The basic chatbot does not require an internet connection, database, external API, or additional software libraries.

7. PROJECT DESCRIPTION

The Basic Chatbot is a simple rule-based Python application that communicates with the user through text input and predefined responses.

The program accepts a message from the user and checks the message against a set of predefined conditions.

For example:

User: hello

Chatbot: Hi!

Similarly:

User: how are you

Chatbot: I'm fine, thanks!

When the user enters:

User: bye

Chatbot: Goodbye!

User: Tell about ai

Chatbot: AI is a fast technology used by evry user to gain knowlge,it helpful for asking questions etc.

The chatbot then ends the conversation.

The chatbot does not generate responses using artificial intelligence. Instead, it follows predefined rules implemented using Python conditional statements.

The program demonstrates how simple conversational behavior can be created using basic programming concepts.

8. FEATURES OF THE CHATBOT

8.1 User Input

The chatbot accepts text messages entered by the user.

8.2 Predefined Responses

The chatbot provides predefined responses for recognized inputs.

8.3 Greeting Response

When the user enters “hello”, the chatbot responds with “Hi!”.

8.4 Status Response

When the user enters “how are you”, the chatbot responds with “I’m fine, thanks!”.

8.5 Goodbye Response

When the user enters “bye”, the chatbot responds with “Goodbye!” and ends the conversation.

8.6 Rule-Based Logic

The chatbot uses predefined rules rather than artificial intelligence.

8.7 Continuous Conversation

A loop allows the chatbot to continue receiving user messages until the user ends the conversation.

8.8 Simple Console Interface

The chatbot uses basic text input and output through the console.

9. METHODOLOGY

The development of the Basic Chatbot was completed through the following steps:

Step 1: Define Requirements

The requirements of the CodeAlpha task were analyzed.

Step 2: Define User Inputs

Basic messages such as “hello”, “how are you”, and “bye” were selected.

Step 3: Define Responses

Predefined responses were created for the recognized inputs.

Step 4: Create the Chatbot Function

A function was used to organize the chatbot's response logic.

Step 5: Accept User Input

The program accepts messages entered by the user.

Step 6: Compare Input

The entered message is compared with predefined conditions using if-elif.

Step 7: Generate Response

The chatbot displays the corresponding predefined response.

Step 8: Continue Conversation

A loop allows the chatbot to continue interacting with the user.

Step 9: End Conversation

When the user enters “bye”, the chatbot displays the goodbye message and terminates the conversation.

10. SYSTEM WORKFLOW

The chatbot follows this basic workflow:

START

↓

Display chatbot welcome message

↓

Ask user for input

↓

Read user message

↓

Check the message

↓

Is input “hello”?

→ Display **“Hi!”**

↓

Is input “how are you”?

→ Display **“I'm fine, thanks!”**

↓

Is input “bye”?

→ Display **“Goodbye!”**

→ End conversation

↓

For other input → Display an appropriate predefined response

↓

Continue conversation

↓

END

11. ALGORITHM

1. Start the program.
2. Define the chatbot function.
3. Display a welcome message.
4. Ask the user to enter a message.

5. Read the user's input.
6. Compare the input with the predefined conditions.
7. If the input is “hello”, display “Hi!”.
8. Else if the input is “how are you”, display “I'm fine, thanks!”.
9. Else if the input is “bye”, display “Goodbye!” and end the conversation.
10. Otherwise, provide a predefined response for unrecognized input.
11. Continue the conversation using a loop.
12. End the program when the user enters “bye”.

12. PYTHON CONCEPTS USED

12.1 If-Elif Statements

if-elif statements are used to compare the user's message with predefined inputs and determine the appropriate response.

12.2 Functions

A function is used to organize the chatbot's response logic.

12.3 Loops

A loop allows the chatbot to repeatedly accept messages from the user until the conversation is ended.

12.4 Input

The input() function is used to receive messages from the user.

12.5 Output

The print() function is used to display chatbot responses.

12.6 Strings

Strings are used to store and compare user messages and chatbot responses.

12.7 Variables

Variables can be used to store the user's input and the chatbot's response.

13. CONVERSATION LOGIC

The chatbot uses a simple rule-based conversation structure.

User Input Chatbot Response

hello Hi!

how are you I'm fine, thanks!

bye Goodbye!

Example Conversation

User: hello

Chatbot: Hi!

User: how are you

Chatbot: I'm fine, thanks!

User: bye

Chatbot: Goodbye!

The chatbot compares the user's input with predefined messages. When a matching input is found, the corresponding response is displayed.

14. IMPLEMENTATION

The Basic Chatbot was implemented using Python's fundamental programming features.

A function is used to organize the chatbot's response logic. The program accepts input from the user and checks the entered message using if-elif conditional statements.

When the user enters a recognized message, the corresponding predefined response is displayed.

A loop is used to keep the conversation running. When the user enters “bye”, the chatbot displays the goodbye response and terminates the conversation.

The implementation follows the simplified requirements provided for the CodeAlpha Basic Chatbot task.

15. TESTING AND RESULTS

Testing was performed using different predefined inputs to verify the chatbot's responses.

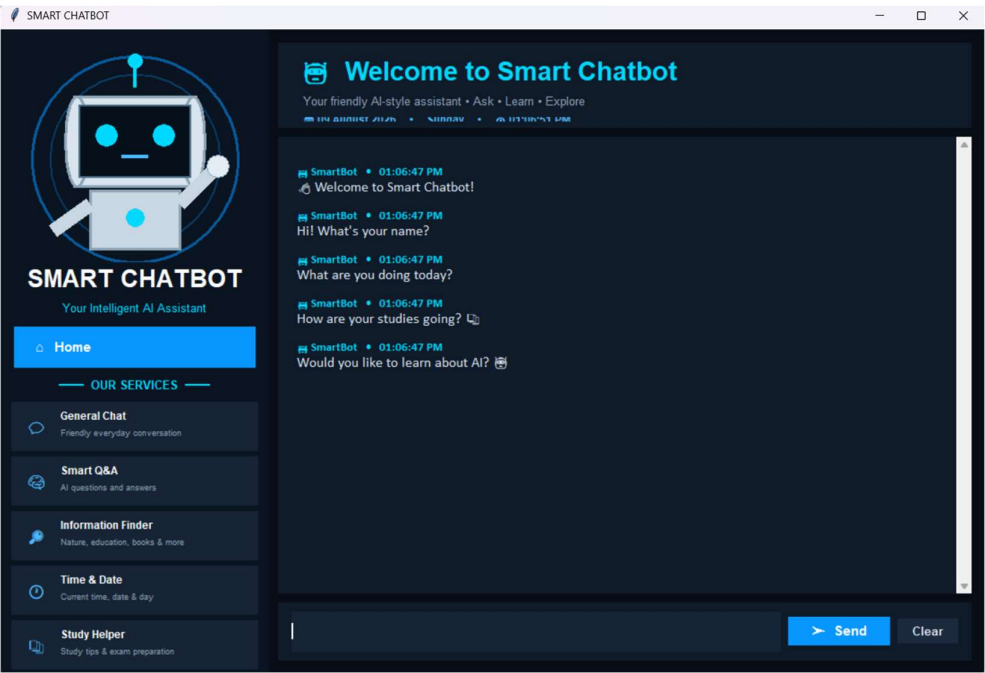
Test Case	User Input	Expected Response	Result
1	hello	Hi!	Pass
2	how are you	I'm fine, thanks!	Pass
3	bye	Goodbye!	Pass
4	Continue conversation	Chatbot continues	Pass
5	Unknown input	Appropriate fallback response	Pass

The testing results indicate that the Basic Chatbot successfully responds to the predefined inputs and follows the expected conversation flow.

16. OUTPUT SCREENSHOTS

16.1 Smart Chatbot Home Screen

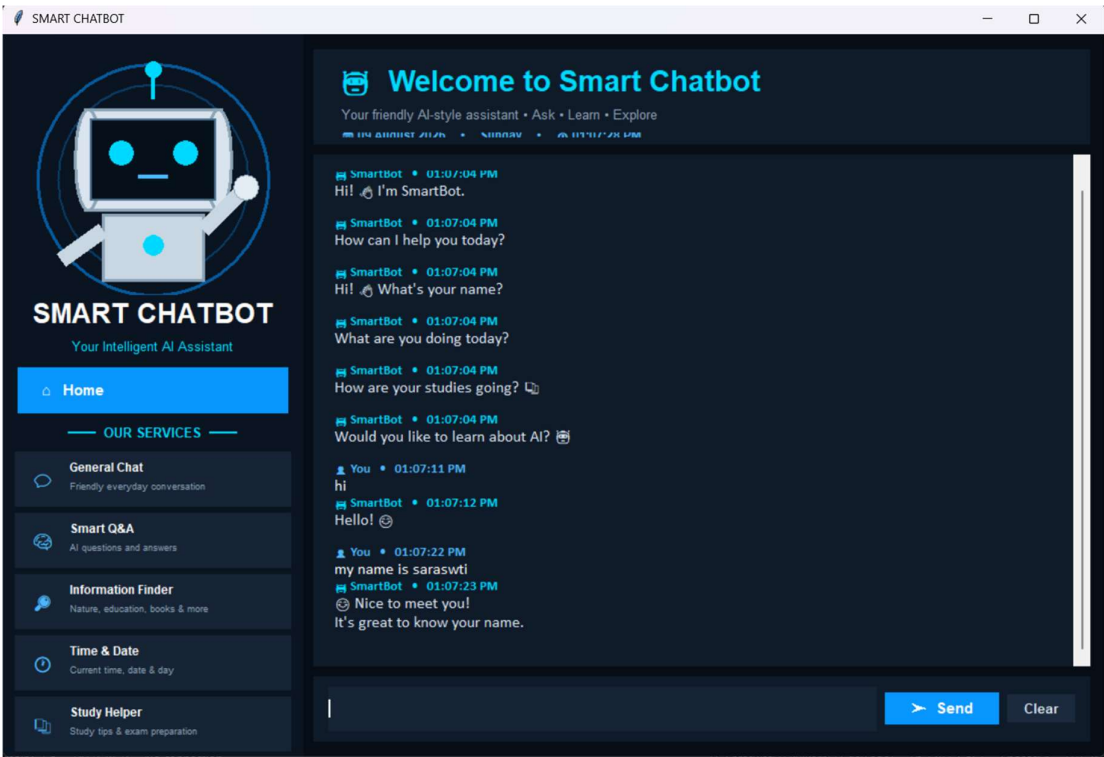
Figure 1: Smart Chatbot Home Interface



The above screenshot shows the main Smart Chatbot interface with the welcome section, chatbot icon, navigation menu, service options, conversation area, and message input box.

16.2 General Chat

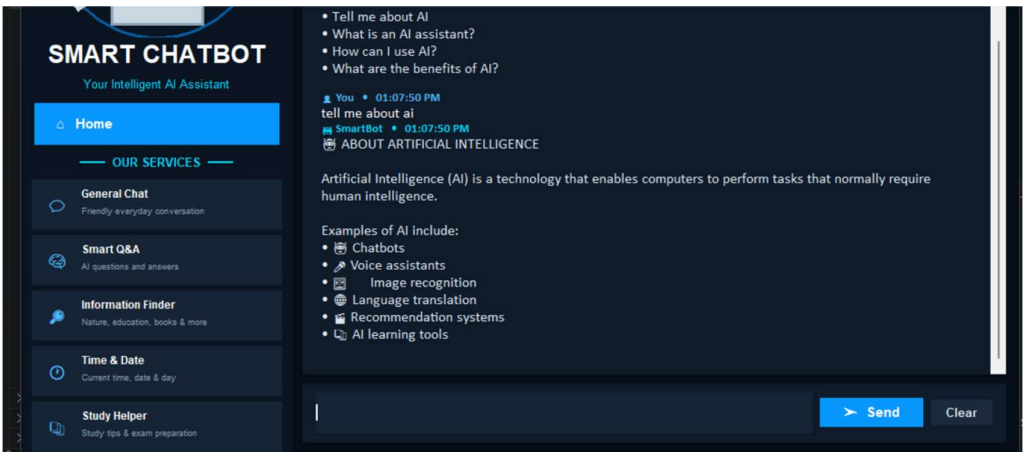
Figure 2: General Chat Feature



This screenshot demonstrates the chatbot responding to a general conversation.

16.3 Smart Q&A

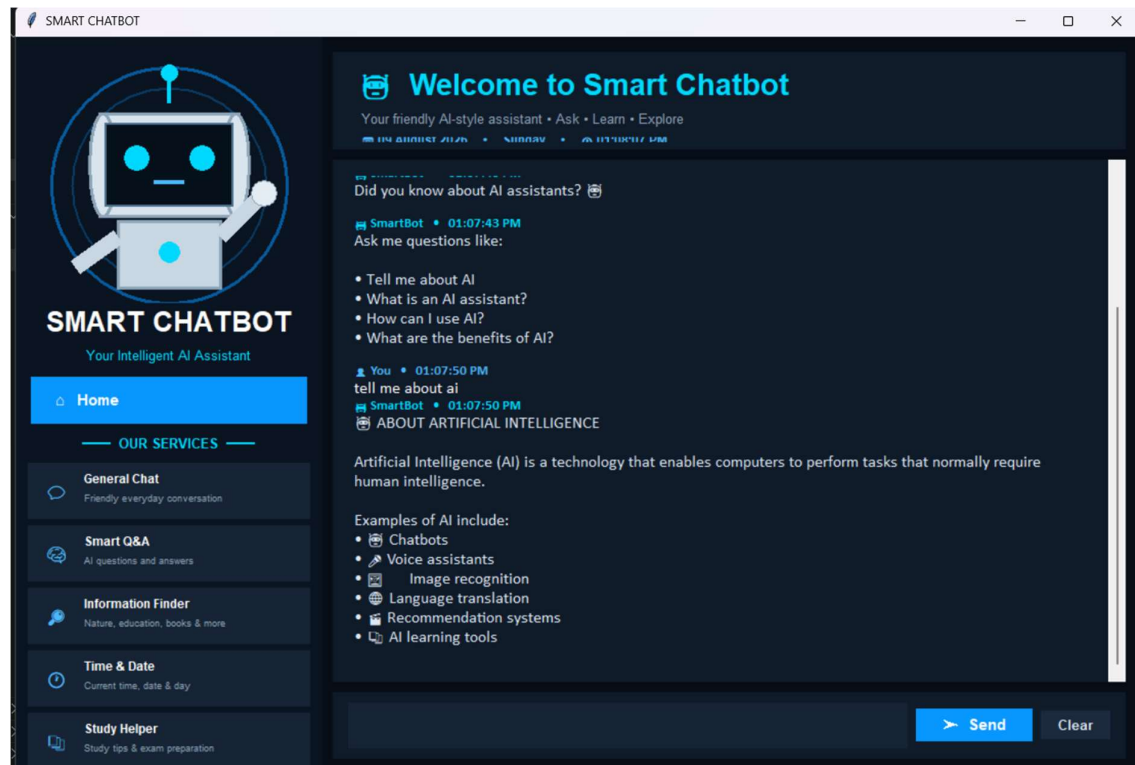
Figure 3: Smart Q&A Feature



This above screenshot demonstrates the chatbot responding to a question using its predefined rules.

16.4 Information Finder

Figure 4: Information Finder Feature

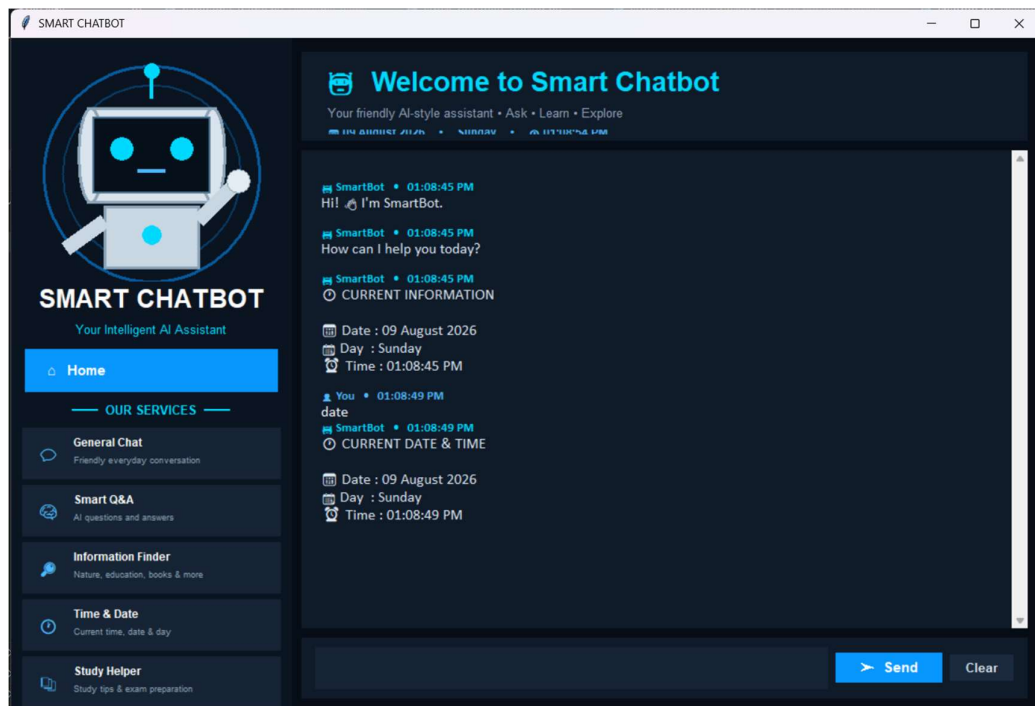


This above screenshot demonstrates the Information Finder functionality.

16.5 Time & Date

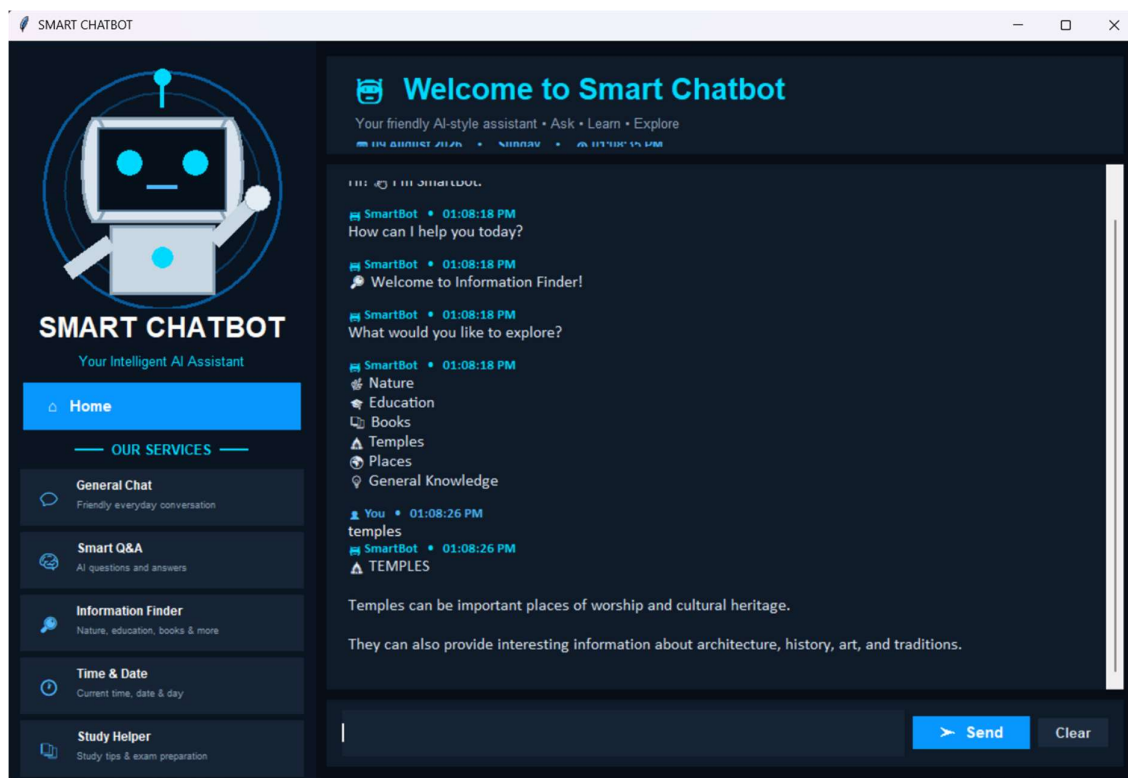
Figure 5: Time & Date Feature

This below screenshot demonstrates the chatbot displaying current time, date, and day information.



16.6 Study Helper

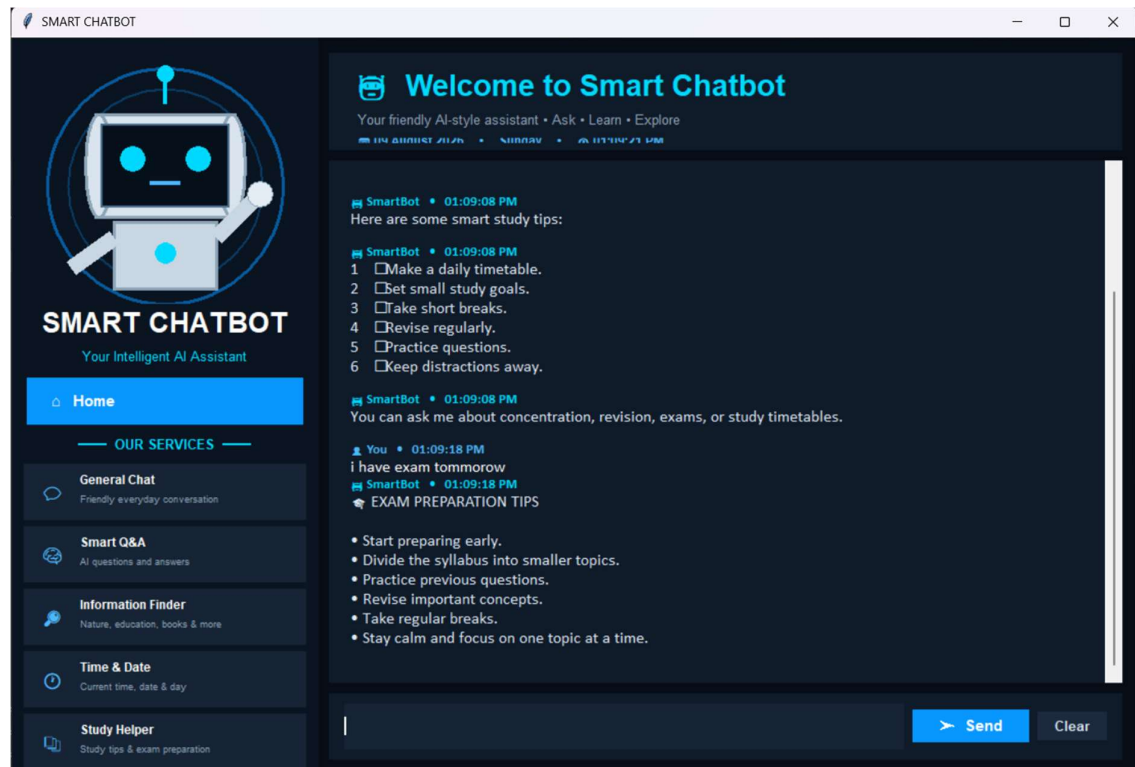
Figure 6: Study Helper Feature



This above screenshot demonstrates the Study Helper providing study and exam preparation tips.

16.7 Exam Preparation Response

Figure 7: Exam Preparation Tips



The above screenshot shows the chatbot responding to the user message **“i have exam tomorrow”** with exam preparation suggestions.

17. ADVANTAGES

The main advantages of the Basic Chatbot are:

- Simple and easy to understand.
- Easy to implement using Python.
- Demonstrates basic conversational logic.
- Provides immediate responses.
- Does not require an internet connection.
- Does not require a database.

- Does not require external APIs.
- Helps beginners understand conditional statements and functions.
- Provides practical experience with loops and user input.
- Can be easily extended with additional responses.

18. LIMITATIONS

The current version of the Basic Chatbot has the following limitations:

1. It supports only predefined inputs.
2. It provides only predefined responses.
3. It does not understand natural language.
4. It does not use artificial intelligence.
5. It does not use machine learning.
6. It cannot understand complex questions.
7. It does not maintain long-term conversation context.
8. It has a limited number of responses.

19. FUTURE ENHANCEMENTS

The Basic Chatbot can be enhanced in several ways in the future.

19.1 More User Inputs

Additional messages and questions can be added.

19.2 More Responses

The chatbot can be expanded with a larger collection of predefined responses.

19.3 Graphical User Interface

A graphical interface could be added to provide a more interactive experience.

19.4 Natural Language Processing

NLP techniques could be introduced to allow the chatbot to understand a wider range of user messages.

19.5 Artificial Intelligence

AI and machine learning techniques could be incorporated in an advanced version.

19.6 Voice Interaction

Voice input and output could be added in a future version.

19.7 Database Integration

A database could be used to store conversation information or chatbot responses.

20. CONCLUSION

The **Basic Chatbot** was successfully developed using Python as part of the **CodeAlpha Python Programming Internship – Task 4**.

The project successfully implements the main requirements of the task by accepting user input and providing predefined responses for messages such as “hello”, “how are you”, and “bye”.

The chatbot demonstrates the practical use of if-elif statements, functions, loops, strings, variables, and input/output operations.

Through this project, practical knowledge of rule-based decision-making and basic conversational program development was gained.

Although the chatbot has a simple scope, it provides a strong foundation for understanding how conversational applications work and how more advanced chatbot features can be developed in the future.

21. REFERENCES

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